

Zarankiewicz for axis-parallel boxes

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Theorem (Erdős–Stone–Simonovits, 1946)

$$\text{ex}(n, H) = \left(1 - \frac{1}{\chi(H) - 1} + o(1)\right) \cdot \binom{n}{2}$$

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- What about bipartite H ? ESS only tells $\text{ex}(n, H) = o(n^2)$!

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- $z(n; t) \stackrel{\text{def}}{=} z(n, n; t, t)$ as shortcut

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Theorem (Bukh, 2024)

$$z(n, n; s, t) = \Omega_t(n^{2-\frac{1}{t}}) \text{ if } s > 10^t$$

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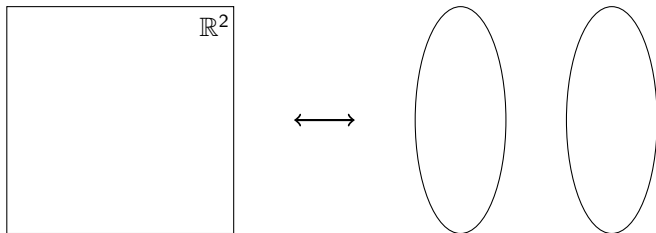
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Theorem (Chen–Liu–Ye, 2025+)

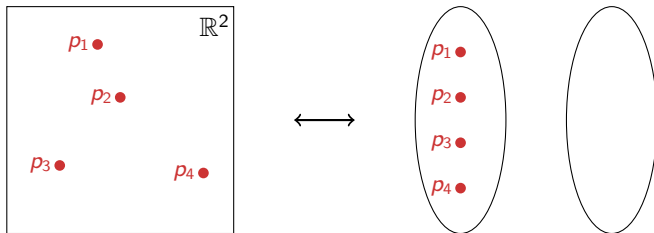
$$z(n, \dots, n; t_1, \dots, t_r) = \Omega_t\left(n^{r - \frac{1}{t_1 \cdots t_{r-1}}}\right) \text{ if } t_r > 10^{t_1 \cdots t_{r-1}}$$

Geometric incidences

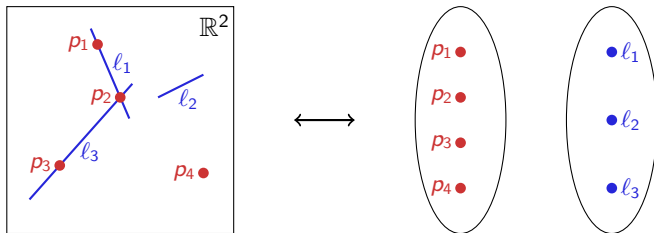
Geometric incidences



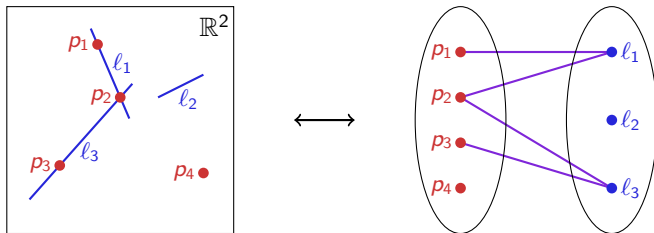
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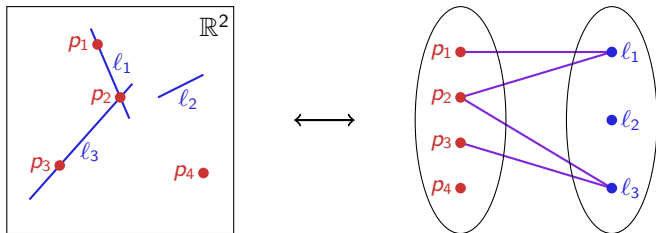
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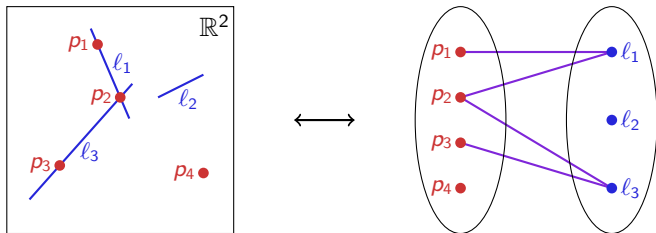


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- point-line incidence graph G is $K_{2,2}$ -free $\xRightarrow{\text{KST}} e(G) = O(n^{3/2})$

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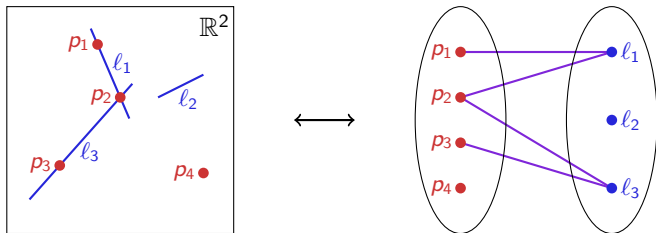


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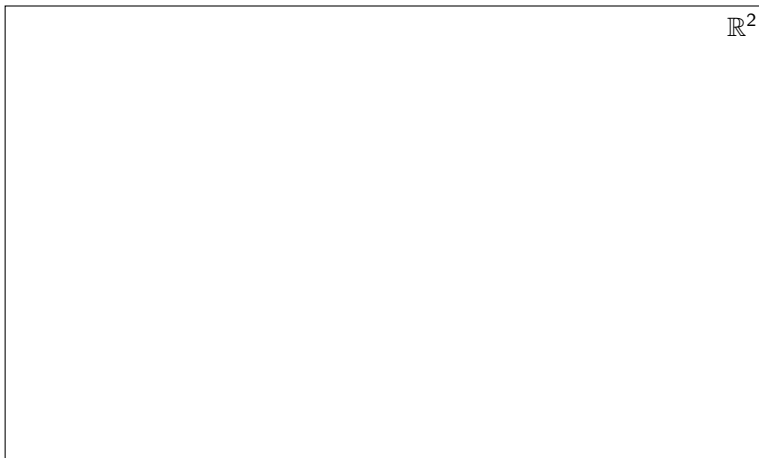
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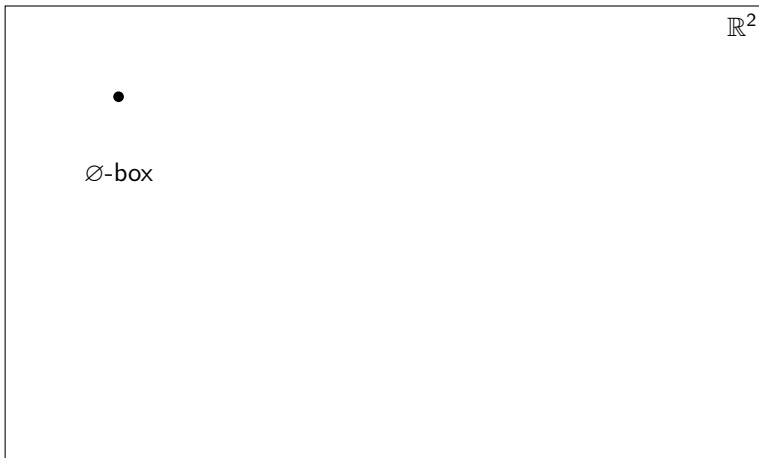
- This $O(n^{4/3})$ bound is tight!

Axis-parallel boxes

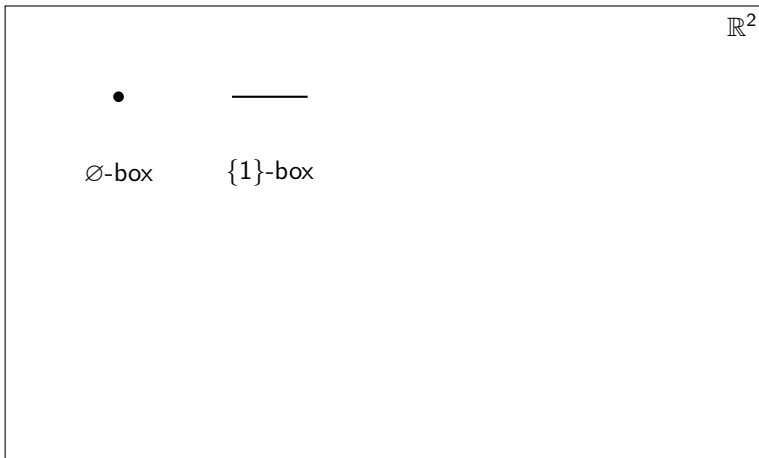
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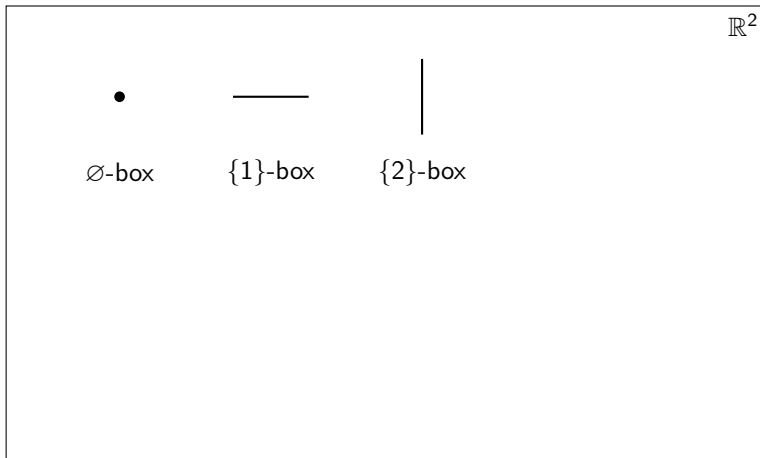
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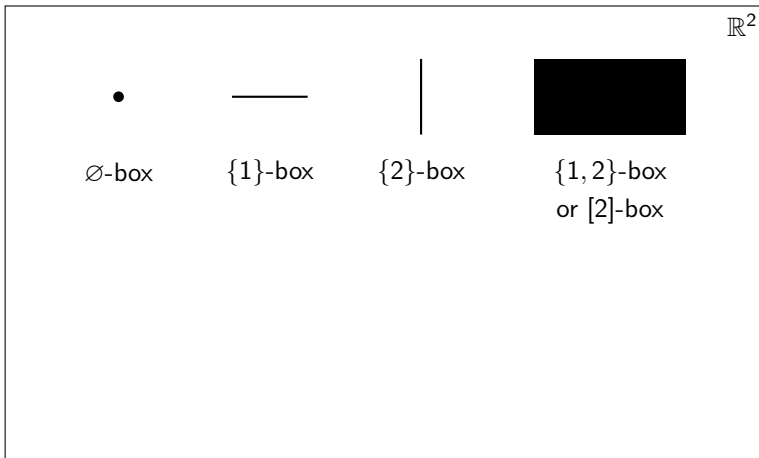
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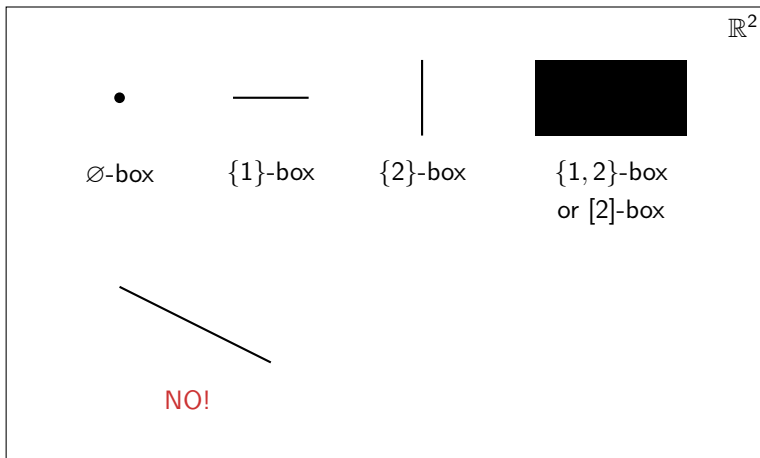
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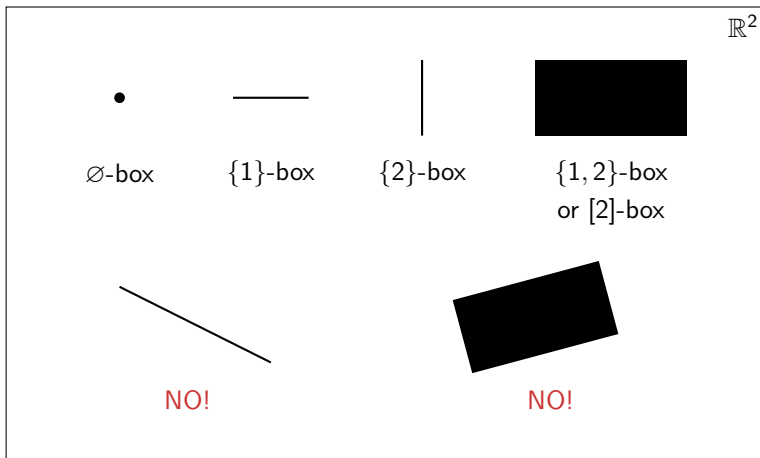
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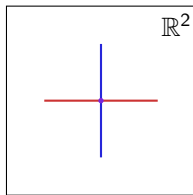
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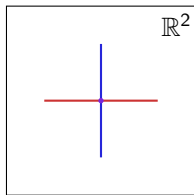
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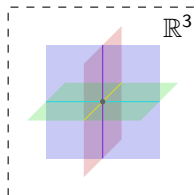
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$$\mathcal{I}_3(\{2, 3\}, \{3, 1\}, \{1, 2\})$$

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$$z_1^{(\emptyset, \{1\})}(n; t) = \Theta(tn) \quad \text{and} \quad z_2^{(\emptyset, [2])}(n; t) = \Theta\left(tn \frac{\log n}{\log \log n}\right)$$

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- when does the extra logarithm factor happen?

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$\vec{F} = (F_1, \dots, F_r)$ in $\mathbb{R}^d$ is *2-coherent* if $\exists k \in [r]$ s.t.

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Theorem (Chao-D.-Liu-Shu-Wang, 2025+)

$$z_d^{\vec{F}}(n; t) = \begin{cases} \Theta_r(tn^{r-1}) & \text{if } \vec{F} \text{ is non-2-coherent,} \\ \Omega\left(tn^{r-1} \frac{\log n}{\log \log n}\right) & \text{if } \vec{F} \text{ is 2-coherent.} \end{cases}$$

Under construction ...